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Aggregation unveiled: A sequential modelling approach to bark beetle outbreaks

Mahdi Salehzadeh *, John M. Stockie, Ailene MacPherson

Department of Mathematics, Simon Fraser University, 8888 University Drive, Burnaby, V5A 1S6, British Columbia, Canada

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Tree-killing bark beetle infestations are a cause of massive coniferous forest mortality impacting forest ecosystems and the ecosystem services they provide. Models predicting bark beetle outbreaks are crucial for forest management and conservation, necessitating studies of the effect of epidemiological traits on the probability and severity of outbreaks. Due to the aggregation behaviour of beetles and host tree defence, this epidemiological interaction is highly non-linear and outbreak behaviour remains poorly understood, motivating questions about when an outbreak can occur, what determines outbreak severity, and how aggregation behaviour modulates these quantities. Here, we apply the principle of distributed delays to create a novel and mathematically tractable model for beetle aggregation in an epidemiological framework. We derive the critical outbreak threshold for the beetle emergence rate, which is a quantity analogous to the basic reproductive ratio, $\mathcal{R}_0$, for epidemics. Beetle aggregation qualitatively impacts outbreak potential from depending on the emergence rate alone in the absence of aggregation to depending on both emergence rate and initial beetle density when aggregation is required. Finally, we use a stochastic model to confirm that our deterministic model predictions are robust in finite populations.

1. Introduction

Tree-killing bark beetles (Scolytinae) represent a prominent disruptive agent within coniferous forest ecosystems across various regions of the world, especially western North America. Economically, the annual cost of insect and pathogen related disturbances is estimated to be 5.7 times larger than that of fire (Dale et al., 2001). Bark beetle outbreaks have triggered extensive, large-scale tree mortality incidents on a global scale (Meddens et al., 2012; Schelhaas et al., 2003). As a result, it is essential to assess and monitor the conditions under which these outbreaks can occur to guide effective environmental policy. Epidemiological models are valuable for understanding outbreak potential and endemic severity, as well as for designing effective control strategies (Garner and Hamilton, 2011; Huppert and Katriel, 2013). While these models are often applied in the context of microparasites, the same approaches can be applied in the context of other parasites and pests, such as tree-killing bark beetles, for which experimental and field studies are challenging. Here, we construct and analyse a novel epidemiological model to investigate bark beetle epidemics, specifically outbreaks of the Mountain Pine Beetle (*Dendroctonus ponderosae*) which are common in the American Pacific Northwest.

Bark beetle infestations exhibit a number of unique epidemiological features. Foremost is the phenomenon of mass attack. In contrast with microparasitic infections, infestations of mass-attacking beetle species (e.g., *Dendroctonus* sp.) are characterized by the cumulative attack of individual host trees by multiple individual beetles. The infestation process begins with an attack phase during which a single female "pioneer" beetle bores through the outer bark into the phloem tissue. She then releases an aggregation pheromone that attracts beetles of both sexes to the tree, where they release their own pheromones. Host trees can resist the attack phase by producing resin (visible externally as pitch tubes) to slow down the boring process or repel the beetles (Amman, 1983). This process continues until the sheer number of beetles overwhelms the host tree defence system at which point the tree is said to be infested. As the attack progresses, beetles produce anti-aggregation pheromones to prevent overexploitation of the limited host tree resources (Logan et al., 1998; Six and Wingfield, 2011). The attack phase is followed by the colonization phase, where beetles build egg galleries, lay eggs, and trees are inoculated with the beetles' fungal mutualists. The eggs then develop into larvae, which build feeding galleries that girdle the tree (Amman and Schmitz, 1988). Tree death often follows as a result of mass attack and the associated colonization of the trees by beetle's fungal mutualist (Zeilinger et al., 2015). After overwintering, the following year the larvae resume development into pupae after

* Corresponding author.
E-mail address: mahdi_salehzadeh@sfu.ca (M. Salehzadeh).

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